

The numbers 1, 2, 3, 4, 5, and 6 are written on slips of paper, and 2 slips are drawn at random one at a time without replacement. Find the given probabilities.

- a. The sum of the numbers is 11.

b. The sum of the number is 7 or less.

c. The first number is 2 or the sum is 5.

a. The probability of an event E, which is a subset of a sample space S of equally likely outcomes, is given by the following, where $n(E)$ and $n(S)$ denotes the size of the given sets.

$$P(E) = \frac{n(E)}{n(S)}$$

Make a table of all the possible outcomes for drawing the two slips at random without replacement.

The table of possible outcomes is shown. The first number represents the number on the first slip of paper drawn, and the second number represents the number on the second slip of paper drawn.

1-2	1-3	1-4	1-5	1-6
2-1	2-3	2-4	2-5	2-6
3-1	3-2	3-4	3-5	3-6
4-1	4-2	4-3	4-5	4-6
5-1	5-2	5-3	5-4	5-6
6-1	6-2	6-3	6-4	6-5

So, to find the probability that the sum of the numbers is 11, first find the sample space for all of the possible sums event.

Using the table, the sample space is 30.

Now, use the table to find the total number of outcomes in which the sum of the two numbers is 11.

The total number of outcomes is 2.

1-2	1-3	1-4	1-5	1-6
2-1	2-3	2-4	2-5	2-6
3-1	3-2	3-4	3-5	3-6
4-1	4-2	4-3	4-5	4-6
5-1	5-2	5-3	5-4	5-6
6-1	6-2	6-3	6-4	6-5

To find the probability of the event, divide the number of outcomes in which the sum of the two numbers is 11 by the sample space. Recall that the sample space is 30.

The probability that the sum of the numbers is 11 is $\frac{1}{15}$.

b. Follow a similar process to find the probability that the sum of the numbers is 7 or less.

Since the sample space has already been determined to be 30, use the table of outcomes to find the number of outcomes in which the sum of the numbers is 7 or less.

The total number of outcomes is 18.

1-2	1-3	1-4	1-5	1-6
2-1	2-3	2-4	2-5	2-6
3-1	3-2	3-4	3-5	3-6
4-1	4-2	4-3	4-5	4-6
5-1	5-2	5-3	5-4	5-6
6-1	6-2	6-3	6-4	6-5

To find the probability of the event, divide the number of outcomes in which the sum is 7 or less by the sample space.

The probability that the sum of the numbers is 7 or less is $\frac{3}{5}$.

c. To find the probability that the first number is 2 or the sum is 5, first identify the correct expression for the probability of the event. Let $P(E)$ represent the probability that the first number is 2 and let $P(F)$ represent the probability that the sum is 5.

The correct expression is $P(E \cup F)$.

Thus, use the Union Rule for Probability, shown in the formula, where E and F are events from a sample space S, to find the probability that the first number is 2 or the sum is 5.

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

First, use the table of possible outcomes to find the probability that the first number is 2, $P(E)$.

$$P(E) = \frac{1}{6}$$

1-2	1-3	1-4	1-5	1-6
2-1	2-3	2-4	2-5	2-6
3-1	3-2	3-4	3-5	3-6
4-1	4-2	4-3	4-5	4-6
5-1	5-2	5-3	5-4	5-6
6-1	6-2	6-3	6-4	6-5

Now, use the table of possible outcomes to find the probability that the sum is 5, $P(F)$.

$$P(F) = \frac{2}{15}$$

1-2	1-3	1-4	1-5	1-6
2-1	2-3	2-4	2-5	2-6
3-1	3-2	3-4	3-5	3-6
4-1	4-2	4-3	4-5	4-6
5-1	5-2	5-3	5-4	5-6
6-1	6-2	6-3	6-4	6-5

Next, find the probability of the intersection of $P(E)$ and $P(F)$. Notice that the probability of the intersection of $P(E)$ and $P(F)$ is the probability that the first number is 2 and the sum is 5.

Use the table of possible outcomes to find the probability that the first number is 2 and the sum is 5. Recall that the sample space is 30.

$$P(E \cap F) = \frac{1}{30}$$

1-2	1-3	1-4	1-5	1-6
2-1	2-3	2-4	2-5	2-6
3-1	3-2	3-4	3-5	3-6
4-1	4-2	4-3	4-5	4-6
5-1	5-2	5-3	5-4	5-6
6-1	6-2	6-3	6-4	6-5

Finally, substitute $\frac{1}{6}$ for $P(E)$, $\frac{2}{15}$ for $P(F)$, and $\frac{1}{30}$ for $P(E \cap F)$ in the Union Rule formula, and solve for the probability that the first number is 2 or the sum is 5, $P(E \cup F)$.

$$\begin{aligned} P(E \cup F) &= P(E) + P(F) - P(E \cap F) \\ &= \left(\frac{1}{6} \right) + \left(\frac{2}{15} \right) - \frac{1}{30} \\ P(E \cup F) &= \frac{4}{15} \end{aligned}$$

The probability that the first number is 2 or the sum is 5 is $\frac{4}{15}$.